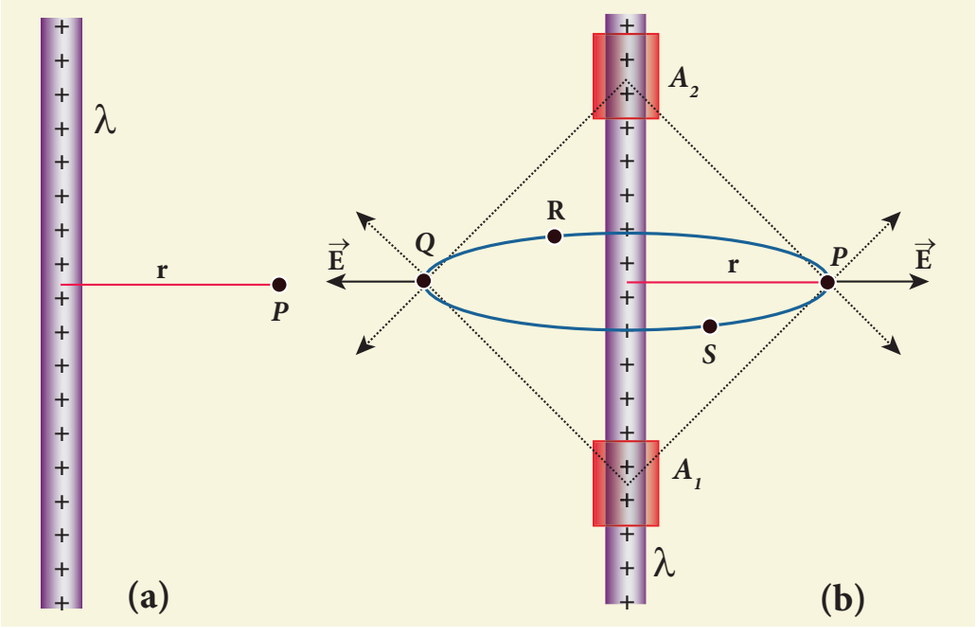


Figure 1.36 Electric field due to infinite long charged wire

a cylindrical symmetry, let us choose a cylindrical Gaussian surface of radius r and length L as shown in the Figure 1.37.

The total electric flux through this closed surface is calculated as follows.

$$\Phi_E = \oint \vec{E}\cdot d\vec{A}$$
$$= \int_{\substack{Curved\\surface}} \vec{E}\cdot d\vec{A} + \int_{\substack{top\\surface}} \vec{E}\cdot d\vec{A} + \int_{\substack{bottom\\surface}} \vec{E}\cdot d\vec{A} \quad (1.63)$$

It is seen from Figure (1.37) that for the curved surface, $\vec{E}$ is parallel to $\vec{A}$ and $\vec{E}\cdot d\vec{A} = E\,dA$. For the top and bottom surfaces, $\vec{E}$ is perpendicular to $\vec{A}$ and $\vec{E}\cdot d\vec{A} = 0$

Substituting these values in the equation (1.63) and applying Gauss law to the cylindrical surface, we have

$$\Phi_E = \int_{\substack{Curved\\surface}} E\,dA = \frac{Q_{encl}}{\epsilon_\circ} \quad (1.64)$$

Since the magnitude of the electric field for the entire curved surface is constant, E is

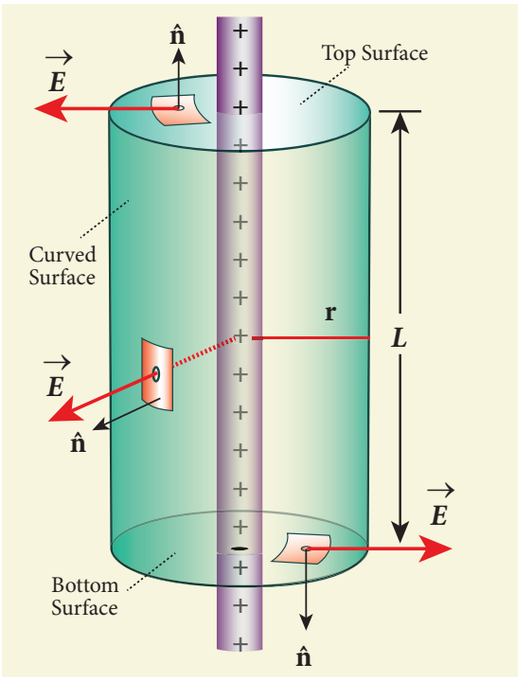


Figure 1.37 Cylindrical Gaussian surface

taken out of the integration and Q_{encl} is given by $Q_{encl} = \lambda L$, where λ is the linear charge density (charge present per unit length).

$$E \int_{\substack{Curved\\surface}} dA = \frac{\lambda L}{\epsilon_\circ} \quad (1.65)$$

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